



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Course Name: College Physics II Dept.: Physics

Exam Duration: 2 hours Exam Paper Setter: Physics Teaching Team

Question No.	1	2	3	4	5	6	7	8	9	10
Score	36	10	12	12	12	12	6			

This exam paper contains 7 questions and the score is 100 in total. (Please hand in your exam paper, answer sheet, and your scrap paper to the proctor when the exam ends.)

Q1. Multiple Choice Questions (3 points each and only one correct answer for each question.)

1~5 C A C C B 6~10 D C D C C 11~12 B D

Long Questions: (Please show the solving process in detail)

Q2. (10 points)

a) Consider a differential element having arc length ds , the (infinitesimal) charge on an element ds of the rod contains charge $dq = \lambda ds$. **1 point**

The element produces a differential electric field at the origin. From the symmetry analysis, we know the electric field sets up by the arc is along the symmetrical axis.

$$dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{r^2} \quad \text{1 point} \quad dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda r d\theta}{r^2} \quad \text{1 point}$$

$$dE_x = \frac{1}{4\pi\epsilon_0} \frac{\lambda r d\theta}{r^2} \cos \theta \quad \text{1 point}$$

$$E_{arc} = \int_{-90^\circ}^{90^\circ} \frac{\lambda R d\theta}{4\pi\epsilon_0 R^2} \cos \theta \quad \text{1 point}$$

$$= \frac{\lambda}{4\pi\epsilon_0 R} (2 \sin 90^\circ) = \frac{\lambda}{2\pi\epsilon_0 R} = \frac{Q}{2\pi^2 \epsilon_0 R^2} = \frac{2kQ}{\pi R^2} = 7.14 \times 10^4 \text{ N/C} \quad \text{2 points}$$

b) $V = \frac{Q}{4\pi\epsilon_0 R} = 2.2 \times 10^3 \text{ V} \quad \text{3 points}$

Q3. (12 points)a) $-5.00 \mu\text{C}$ (inner surface), $-3.00 \mu\text{C}$ (outer surface) **4 points**

$$b) \begin{cases} r > R_3 & E = \frac{q_{out}}{4\pi\epsilon_0 r^2} \\ R_2 < r < R_3 & E = 0 \\ R_1 < r < R_2 & E = \frac{q}{4\pi\epsilon_0 r^2} \\ r < R_1 & E = 0 \end{cases}$$

$$r = 2R_3 = 60\text{cm} \Rightarrow E = \frac{q_{out}}{4\pi\epsilon_0 r^2} = \frac{-3 \times 10^{-6}}{4\pi \times 8.85 \times 10^{-12} \times 0.6^2} = -7.5 \times 10^4 \text{ N/C} \quad \text{3 points}$$

$$r = R_1 / 2 \Rightarrow V = \frac{q_{out}}{4\pi\epsilon_0 R_3} + \frac{q}{4\pi\epsilon_0 R_1} - \frac{q}{4\pi\epsilon_0 R_2}$$

$$= \frac{-3 \times 10^{-6}}{4\pi \times 8.85 \times 10^{-12} \times 0.2} + \frac{5 \times 10^{-6}}{4\pi \times 8.85 \times 10^{-12} \times 0.05} - \frac{5 \times 10^{-6}}{4\pi \times 8.85 \times 10^{-12} \times 0.15} = 4.6 \times 10^5 \text{ V}$$

3 points

c) The location of the inner sphere does not affect the charge distribution on the outer surface of the conducting shell, so does the electric field for the region $r > R_3$. And the conducting shell is a equipotential body, so the potential at the inner surface is exactly the potential at out surface, so we have:

$$V = \frac{q_{out}}{4\pi\epsilon_0 R_3} = \frac{-3 \times 10^{-6}}{4\pi \times 8.85 \times 10^{-12} \times 0.2} = -1.3 \times 10^5 \text{ V} \quad \text{2 points}$$

Q4. (12 points)

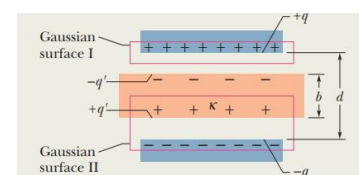
$$a) C_0 = \frac{\epsilon_0 A}{d} = \frac{(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(0.12 \text{ m}^2)}{1.2 \times 10^{-3} \text{ m}} = 0.885 \text{ nF.} \quad \text{2 points}$$

$$q = C_0 V = (0.885 \text{ nF})(100 \text{ V}) = 88.5 \text{ nC.} \quad \text{2 points}$$

(b) Assume the charge on the positive plate is q and then the electric field in the region between the vacuum and the dielectric and the electric field inside the dielectric can be got from Gauss' law.

From the Gaussian surface II, we have

$$\text{Gaussian surface II: } \kappa EA = \frac{q}{\epsilon_0} \Rightarrow E_2 = \frac{q}{\kappa A \epsilon_0} \quad \text{3 points}$$



$$E_2 = \frac{q}{\kappa A \epsilon_0} = \frac{88.5 \times 10^{-9}}{20 \times 0.12 \times 8.85 \times 10^{-12}} = 4.16 \times 10^3 \text{ V/m}$$

1 point

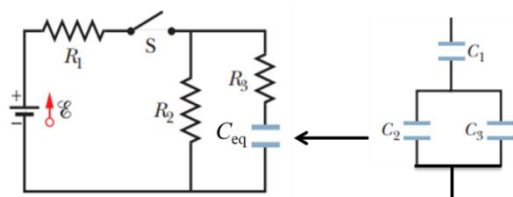
c) The potential difference between the two plates is:

$$V = E_1(d - b) + E_2b$$

And the capacitance of the capacitor is given by the definition:

$$C = \frac{q}{V} = \frac{q}{E_1(d - b) + E_2b} = \frac{q}{\frac{q}{A\epsilon_0}(d - b) + \frac{q}{\kappa A\epsilon_0}b} = \frac{\kappa A\epsilon_0}{\kappa(d - b) + b} \quad 3 \text{ points}$$

$$C = \frac{(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(0.12 \text{ m}^2)(20)}{(20)(1.2 - 0.40) \times 10^{-3} \text{ m} + 0.4 \times 10^{-3} \text{ m}} = 1.30 \text{ nF} \quad 1 \text{ point}$$

Q5. (12 points)**Solution:**

$$a) \quad \frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2 + C_3} = \frac{1}{C} + \frac{1}{2C} \Rightarrow C_{eq} = \frac{2}{3}C = 4.0 \text{ } \mu\text{F} \quad 3 \text{ points}$$

b) When the capacitors are fully charged, there is no current in equivalent capacitor, so the potential difference across C_{eq} is the potential difference across R_2

$$V_{eq} = R_2 \frac{\epsilon}{R_1 + R_2} = \frac{\epsilon}{2} \quad 2 \text{ points}$$

The charge on C_{eq} is

$$Q_{eq} = C_{eq} V_{eq} = \left(\frac{2}{3}C\right)\left(\frac{\epsilon}{2}\right) = \frac{C\epsilon}{3} = 4.0 \text{ mC} \quad 3 \text{ points}$$

c) From the combination of the capacitors, we know $q_2 = q_{eq}/2$ 1 point

For the discharging circuit, we have:

$$q_2 = \frac{q_{eq}}{2} = \frac{1}{2} Q_{eq} e^{-t/2R\frac{2C}{3}} = \frac{C\epsilon}{6} e^{-3t/4RC} = 2e^{-t/8} \text{ mC} \quad 3 \text{ points}$$

Q6. (12 points)

a) We choose the concentric circle as the Ampere's loop:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc} \Rightarrow B 2\pi r = \mu_0 I \frac{\pi r^2}{\pi R^2} \Rightarrow B = \frac{\mu_0 I r}{2\pi R^2} \quad r < R \quad \text{3 points}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc} \Rightarrow B 2\pi r = \mu_0 I \Rightarrow B = \frac{\mu_0 I}{2\pi r} \quad r > R \quad \text{4 points}$$

Ampere's loop 1 point

$$\text{b) } \oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc} \Rightarrow B 2\pi r = \mu_0 \left(I + \int_a^r \frac{A}{r} 2\pi r dr \right) \quad \text{3 points}$$

$$\Rightarrow B = \frac{\mu_0 I}{2\pi r} + \frac{\mu_0 2\pi A(r-a)}{2\pi r} = \text{constant} \Rightarrow A = \frac{I}{2\pi a} \quad \text{0.5 point}$$

Positive A indicates that the current density is out of paper 0.5 point

Q7. Blank-filling Questions. (Complete each of the following questions according to your calculation)

1. (3 points).

$$V_{z=0} = \int_A dV = \int_A \frac{\sigma dA}{4\pi\epsilon_0 R} = \frac{\sigma 2\pi R^2}{4\pi\epsilon_0 R} = \frac{\sigma R}{2\epsilon_0}$$

$$V_{z=R} = \int_A dV = \int_0^{\pi/2} \frac{2\pi R^2 \sigma \sin\theta d\theta}{4\pi\epsilon_0 (R^2 + 2R^2 \cos\theta + R^2)^{1/2}} = -\frac{2\pi\sigma d(2R^2 + 2R^2 \cos\theta)}{8\pi\epsilon_0 (2R^2 + 2R^2 \cos\theta)^{1/2}}$$

$$= -\frac{\sigma}{2\epsilon_0} (2R^2 + 2R^2 \cos\theta)^{1/2} \Big|_0^{\pi/2} = \frac{\sigma R}{2\epsilon_0} (2 - \sqrt{2})$$

$$\Delta K = 0 \Rightarrow \Delta U = 0 \Rightarrow \Delta V = 0 \Rightarrow \frac{\sigma R}{2\epsilon_0} - \frac{\rho d}{2\epsilon_0}(R) = \frac{\sigma R}{2\epsilon_0} (2 - \sqrt{2})$$

$$\Rightarrow \sigma = \frac{\rho d(R)}{(\sqrt{2}-1)R} = \frac{R\rho}{4(\sqrt{2}-1)} = 0.60\rho R$$

3. (3 points).

$$\Delta V = V_{\text{outer}} - V_{\text{inner}} = -\int_a^b \frac{\sigma_{\text{in}} a}{\epsilon_0 r} dr = -\frac{\sigma_{\text{in}} a}{\epsilon_0} \ln \frac{b}{a}$$

$$\Rightarrow \sigma_{\text{in}} = -\frac{\epsilon_0 \Delta V}{a \ln 2} = -2.55 \mu\text{C}/\text{m}^2$$

$$i = \frac{q}{T} = \frac{\sigma 2\pi r L}{2\pi / \omega} = \sigma \omega r L$$

$$BL = \mu_0 i_{\text{enc}} = \mu_0 (\sigma_{\text{in}} \omega a L + \sigma_{\text{out}} \omega b L) \Rightarrow B = \mu_0 (\sigma_{\text{in}} \omega a + \sigma_{\text{out}} \omega b) = \mu_0 \omega \left(-\frac{\epsilon_0 \Delta V}{\ln 2} + \sigma_{\text{out}} b \right)$$

$$= 6.5 \times 10^{-11} \text{ T}$$

$$6.5 \pm 0.1 \times 10^{-11} \text{ T}$$